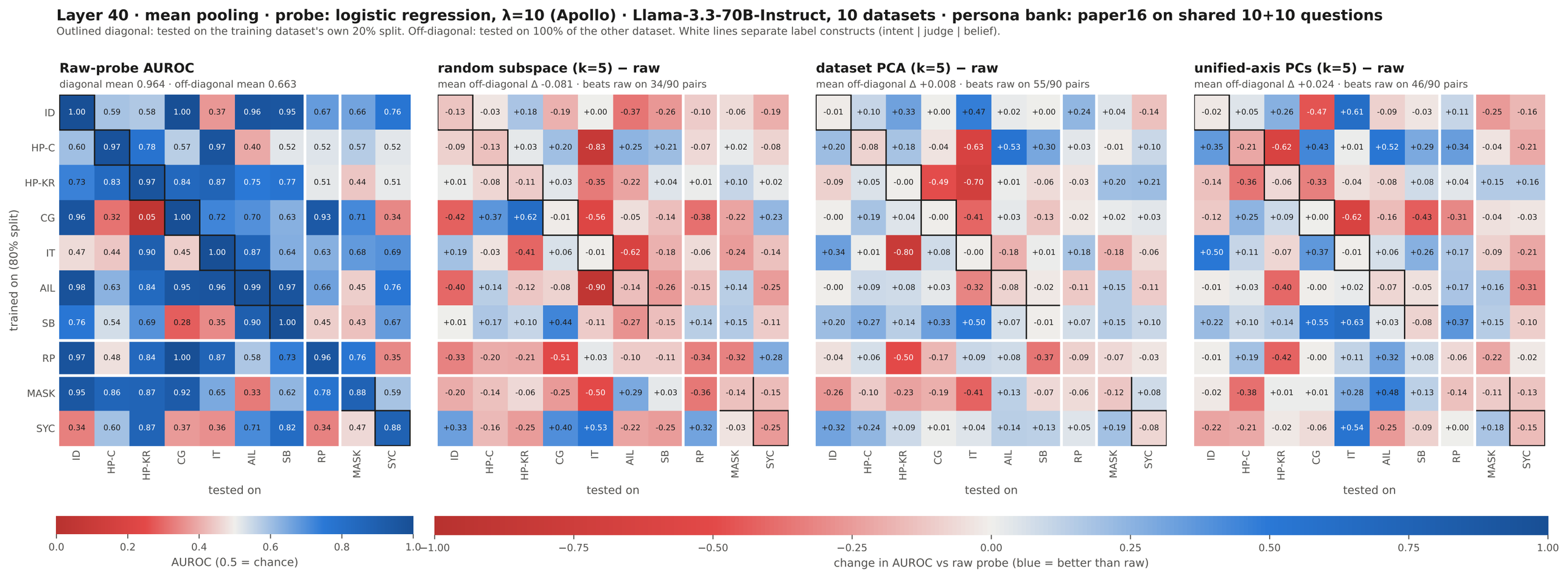

Layer 40 · mean pooling · probe: logistic regression, λ=10 (Apollo) · Llama-3.3-70B-Instruct, 10 datasets · persona bank: paper16 on shared 10+10 questions
Outlined diagonal: tested on the training dataset's own 20% split. Off-diagonal: tested on 100% of the other dataset. White lines separate label constructs (intent | judge | belief).
Raw-probe AUROC
diagonal mean 0.964 · off-diagonal mean 0.663
random subspace (k=5) − raw
mean off-diagonal Δ -0.081 · beats raw on 34/90 pairs
dataset PCA (k=5) − raw
mean off-diagonal Δ +0.008 · beats raw on 55/90 pairs
unified-axis PCs (k=5) − raw
mean off-diagonal Δ +0.024 · beats raw on 46/90 pairs
trained on (80% split)
ID HP-C HP-KR CG IT AIL SB RP MASK SYC
tested on
AUROC (0.5 = chance)
change in AUROC vs raw probe (blue = better than raw)